

RENIN-ANGIOTENSIN SYSTEM (RAS)

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- **Primary Substrate** : α_2 -Globulin (**Angiotensinogen**), synthesized mainly by the liver, serves as the primary substrate for the enzyme **renin** (secreted by the juxtaglomerular apparatus of the kidneys).

PEPTIDE FORMS OF ANGIOTENSIN

Peptide Form	Chain Length	Biological Activity & Key Properties
Angiotensin I	Decapeptide (10 aa)	• Physiologically inactive ; serves solely as an intermediate precursor.
Angiotensin II	Octapeptide (8 aa)	• Most active form of the system. • Very short half-life: 1 – 2 minutes.
Angiotensin III	Heptapeptide (7 aa)	• Active derivative; stimulates aldosterone secretion and causes vasoconstriction.

Peptide Form	Chain Length	Biological Activity & Key Properties
Angiotensin IV	Hexapeptide (6 aa)	• Further cleavage product; possesses moderate biological activity.

KEY ENZYMES

1. Angiotensin-Converting Enzyme (ACE) :



- Secreted predominantly by vascular endothelial cells of the **lungs** (and renal capillaries).

2. Angiotensinases :



- Present in **RBCs** and the **vascular beds** of many peripheral tissues.

SYSTEM CASCADE & REGULATION

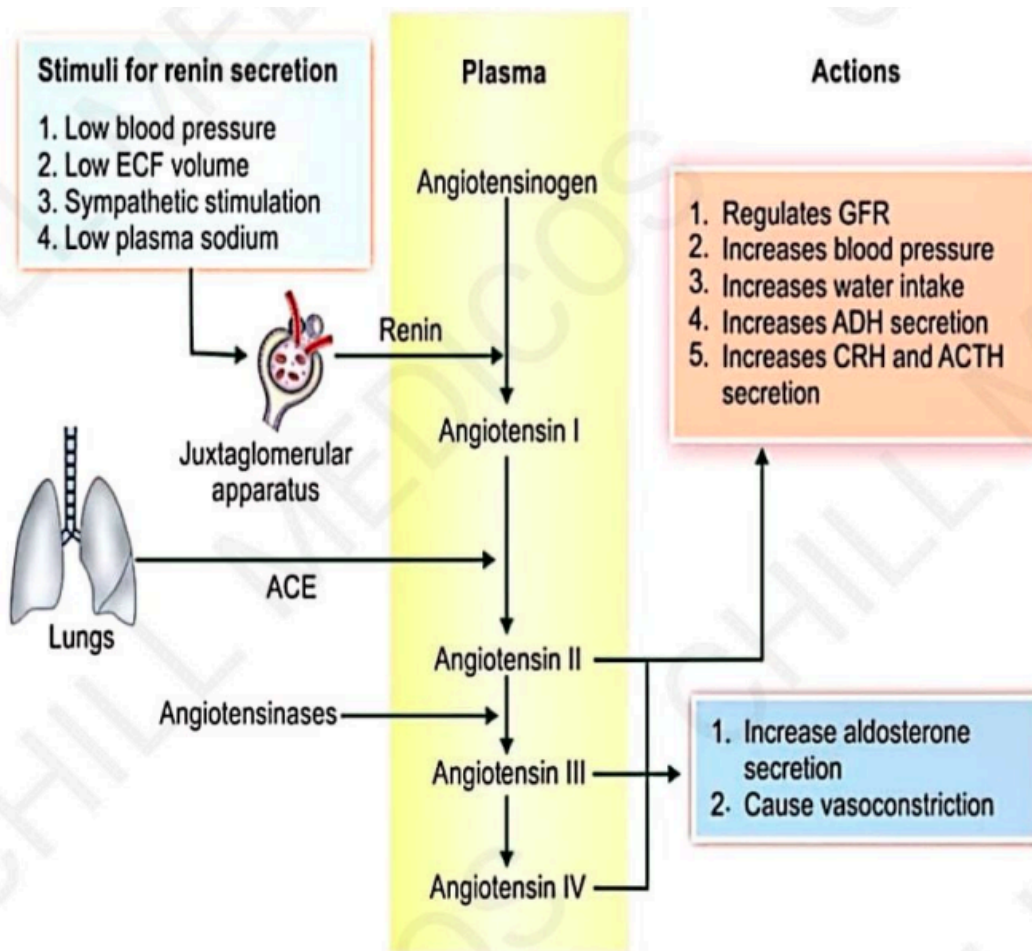
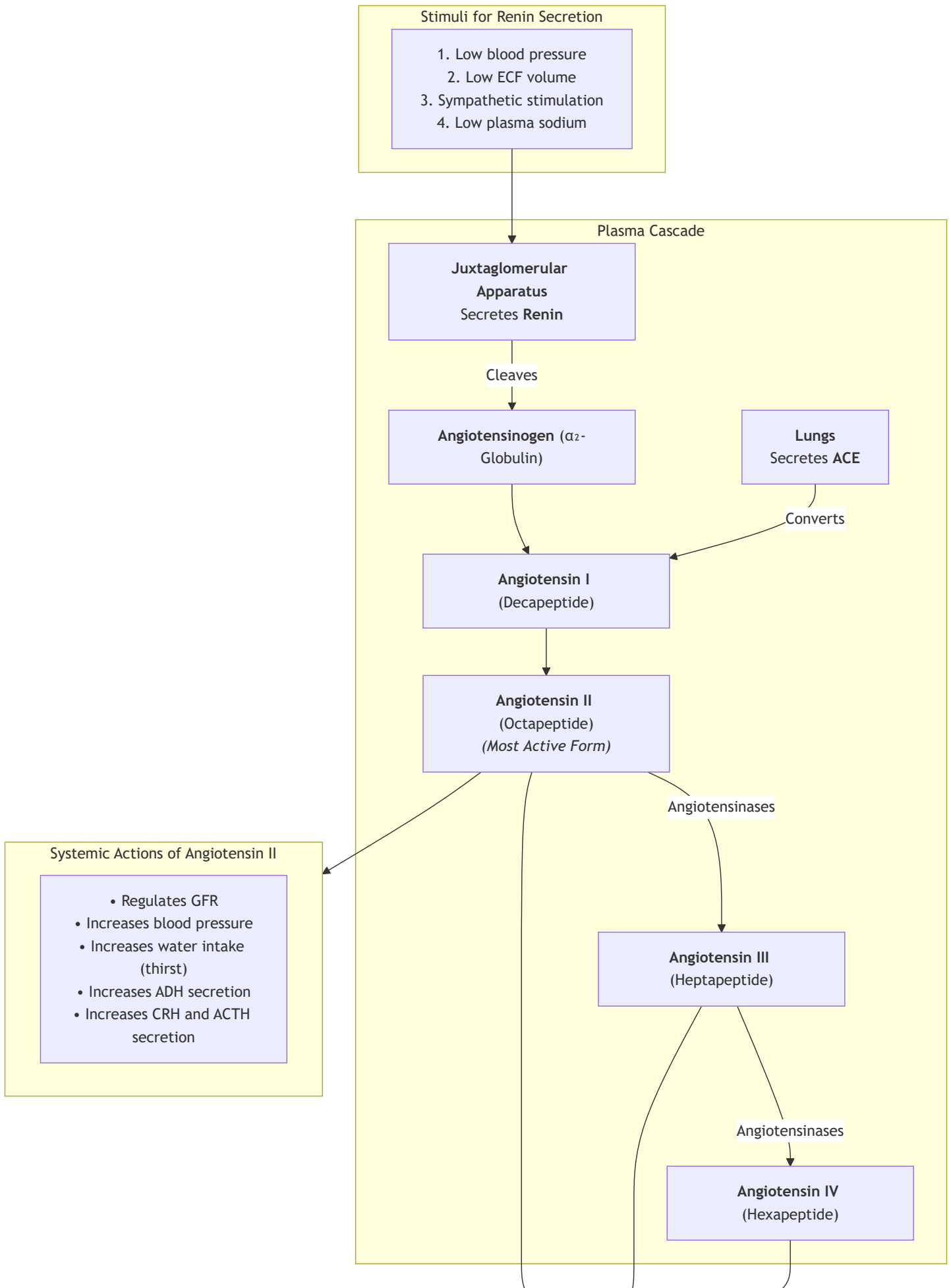
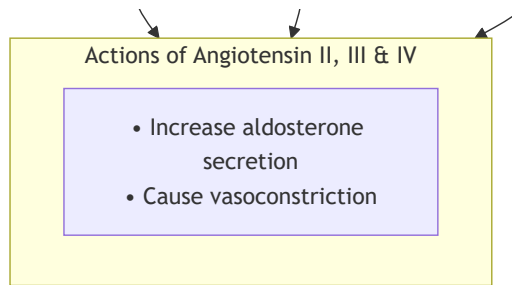


FIGURE 50.2: Renin-angiotensin system. ECF = Extracellular fluid, ACE = Angiotensin-converting enzyme, GFR = Glomerular filtration rate, ADH = Antidiuretic hormone, CRH = Corticotropin-releasing hormone, ACTH = Adrenocorticotrophic hormone.





ACTIONS OF ANGIOTENSINS

1. ANGIOTENSIN I

- **Physiologically Inactive** : Serves only as a precursor molecule for Angiotensin II.

2. ANGIOTENSIN II (*Most Active Form*)

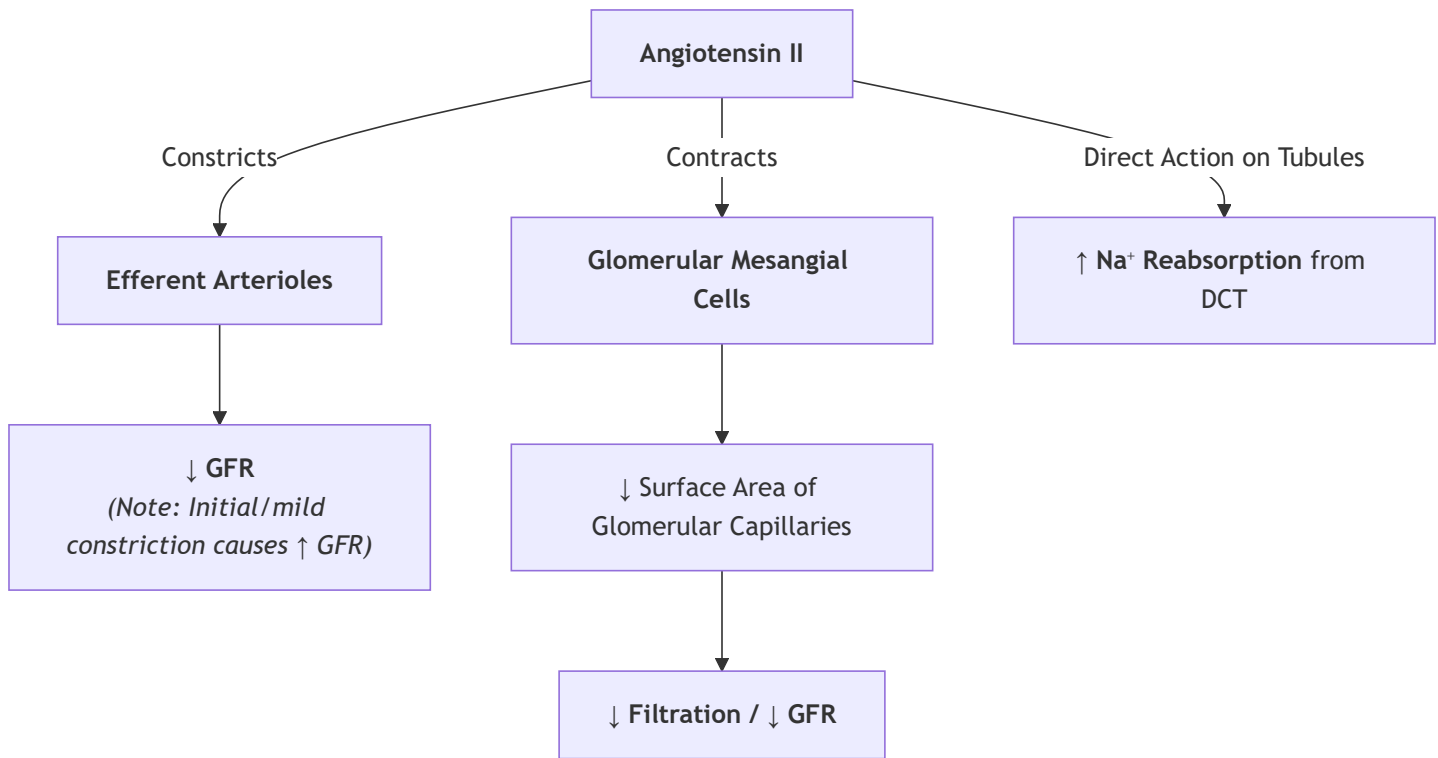
[A] Actions on Blood Vessels

- **Direct Vasoconstriction** : Potent direct arteriolar vasoconstrictor $\longrightarrow$ $\uparrow$ increases total peripheral resistance and $\uparrow$ **raises Blood Pressure (BP)** (*hypertensin action*).
- **Indirect BP Increase** : Stimulates the release of **noradrenaline** (a potent vasoconstrictor) from postganglionic sympathetic nerve fibers.

[B] Actions on Adrenal Cortex

- Stimulates the **Zona Glomerulosa** cells of the adrenal cortex to secrete **Aldosterone**.
- Aldosterone acts on the **Distal Convolute Tubules (DCT)** of kidneys $\longrightarrow$ $\uparrow$ increases Na^+ retention (and secondary water retention) $\longrightarrow$ $\uparrow$ Blood Pressure.

[C] Actions on Kidney & GFR Regulation



[D] Actions on Brain / Central Nervous System

- **Inhibits Baroreceptor Reflex :** Blunts the reflex response that normally opposes elevations in BP, thereby sustaining hypertension.
- **Stimulates Thirst Center :** ↑ Increases water intake (dipsogenic effect).
- **↑ Increases CRH Secretion :** Acts on the hypothalamus to release Corticotropin-Releasing Hormone (CRH) → ↑ stimulates **ACTH secretion** from the anterior pituitary.
- **↑ Increases ADH Secretion :** Stimulates the hypothalamus to release **Antidiuretic Hormone (ADH / Vasopressin)**, promoting renal water reabsorption.

3. ANGIOTENSIN III

- **↑ Increases Blood Pressure.**
- **Potency Profile :**
- **100% adrenocortical stimulating activity** compared to Angiotensin II (equally potent at stimulating aldosterone release).
- **40% vasopressor (vasoconstrictor) activity** of Angiotensin II.

4. ANGIOTENSIN IV

- Possesses moderate **adrenocortical stimulating** and **vasopressor activities**.